



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 28.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **EEE G614**
Course Title : **Advanced Wireless Communications**
Instructor-in-Charge : **Dr. Monolina Dutta**
Instructor(s) : **Dr. Monolina Dutta**

1. Course Description:

Evolution of wireless cellular technologies to 5G and beyond, review of matrix and signal theory for communication applications, introduction to stochastic geometry for performance analysis of wireless networks, D2D communications- modeling and analysis, cooperative communications-buffer-aided relaying and performance analysis, modulation and multiple access techniques-NOMA, OTFS, OAM. Introduction to potential technologies for beyond 5G/6G communications: IRS, molecular communications, and AI in wireless communications.

2. Scope and Objective of the Course:

- This course aims to introduce the students to the advances in wireless communications and expose them to the research options available for applications in the real-world implementation of the next-generation wireless networks.
- The course will introduce the students to current trends in the wireless industry with a specific focus on the 5G and beyond 5G communication networks.
- The course aims to equip students with tools for designing the engineering aspects of wireless communications.
- New techniques envisaged to be a part of the next-generation wireless networks, like D2D communications, NOMA, OTFS, and stochastic geometry applied to wireless communications, to name a few, will be covered to make the student industry and research-ready.

3. Course Learning Outcomes:

- Evaluate the trade-offs in system design parameters (e.g., power, bandwidth, latency, mobility) in the context of modern wireless networks.
- Design and evaluate advanced modulation, coding, and diversity techniques for improving link reliability and spectral efficiency in various wireless environments.
- Analyze complex wireless channel models (e.g., time-varying fading, spatially correlated MIMO channels) and assess their implications on communication system performance.

4. Text Books:

- Jonathan Rodriguez, "Fundamentals of 5G Mobile Networks", Wiley, 2015.
- Bruno Clerckx and Claude Oestges, MIMO Wireless Networks: Channels, Techniques and Standards for Multi-Antenna, Multi-User and Multi-Cell Systems, Academic Press (Elsevier), UK, Jan 2013.

Reference Books:



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

1. Erik G. Larsson, Petre Stoica, and Girish Ganesan. 2003. Space-Time Block Coding for Wireless Communications. Cambridge University Press, USA.
2. A. Goldsmith, Wireless Communications, Cambridge Univ. Press, 2005.
3. M. Haenggi, Stochastic Geometry for Wireless Networks. New York, NY, USA: Cambridge Univ. Press, 2013.
4. C. Johnson, 5G New Radio in Bullets. Independent Publishing, ISBN: 978-1081444594, 2019.
5. D. Stoyan, W. Kendall, and J. Mecke, Stochastic Geometry and Its Applications, 2nd ed., New York, NY, USA: Wiley, 1987.
6. Research papers (to be announced in the class)

5. Course will be conducted through weekly lecture sessions (3hrs/week), which will make extensive use of Blackboard and research papers based exercises over a period of 14 weeks covering the semester.

6. Course Plan:

Module	Lecture Session	Coverage	Ref.	Learning outcomes
Introduction	1	Evolution and journey of wireless technologies to 5G and beyond 5G	T1, research papers, and class notes	This module will introduce the key next-generation wireless technologies, the challenges, needs, and evolution
	2-3	5G networks: 5G NR characteristics, architecture, performance metrics, key technologies, standardization activities, and introduction to 6G		
	4-5	Spectrum and regulations- spectrum challenges in 5G and beyond 5G-mmWave and THz frequency band for wireless communications		
Mathematical tools for wireless network analysis	6	Signal Theory- Introduction	T2 and class notes	This module will briefly review the matrix and signal theory. Further, it will introduce stochastic geometry
	7-10	Characteristic function, PDF, CDF, MGF, Moments of random variables-discrete and continuous random variables		
	11-18	Gaussian random variable, complex Gaussian, Rayleigh, Rician, Chi-square, Gamma, exponential, and Nakagami-m distributions		
	19-22	Stochastic Processes-White and Gaussian		
	23-26	A review of matrix theory for applications to the wireless network design		



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

D2D communications, modulation, and multiple access schemes	27- 32 33-35	D2D communications-modeling and analysis NOMA; OTFS, OAM, and comparisons with OFDM	Research papers, class notes	Evolution of D2D networks, modulation and coding schemes
Emerging trends	36-40	Introduction to other potential technologies with applications in next-generation communication networks: Error correction coding schemes, IRS, AI applications to the wireless communication networks	Class notes and research papers	New trends in mobile networks

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	20	8/10/2026 AN2	Closed Book
Seminar & Attendance	--	20	To be announced in class	Open Book
Take-home assignments	--	20	To be announced in class	Open Book
Quiz (s)	--	10	To be announced in class	Closed Book
Comprehensive Examination	180 Min.	30	12 /12/2026 AN	Closed Book

8. Chamber Consultation Hour: To be announced in the lecture.

9. Notices: All notices will be posted on Nalanda/announced in the lecture.

10. Make-up Policy: No make-up will be given for quizzes; however, for other components, make-up will be given ONLY in cases of sickness (hospitalization). In such cases, students must produce sufficient proof and pre-inform the Instructor.

11. Note (if any): Strict action, along with awarding of zero marks, will be taken in case of plagiarism of the lab reports. All submissions will be checked by a plagiarism detection tool.

Instructor-in-charge
Course No. EEE G614